



TOYJOY

Inventory Planning & Analytics

Toy Inventory Optimization Analysis

Reducing Overstock, Preventing Stockouts & Improving Cash Flow

FY 2023 - 2025 · Retail & Online Channels · Seasonal Demand & Reorder Strategy

Rs 38.58 Cr

Total Net Revenue

9.96 Lakh

Units Sold (3 Yrs)

54.3%

Blended Profit Margin

12

Stores Analyzed

FRAMING

Project Objectives & Scope

ToyJoy sells through retail stores and online channels. After last holiday season, management saw excess inventory of some products and stockouts of others — this analysis reduces overstock, prevents lost sales, and frees up cash.



Cut Excess Inventory

Identify slow-moving products tying up ~Rs 2.26 Cr in unsold stock and right-size reorder quantities.



Prevent Stockouts

Fix the 6 high-demand products stocking out ~17% of weeks; protect them from lost sales during peak season.



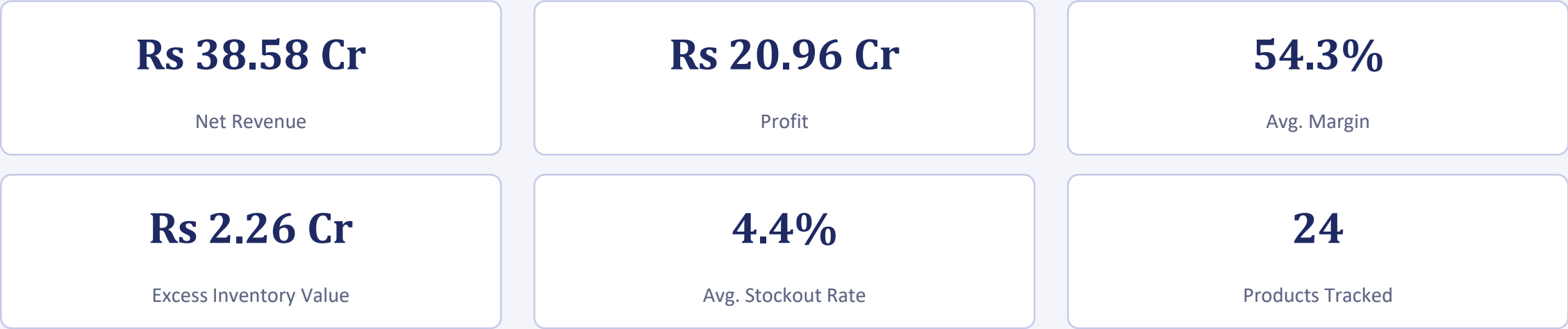
Forecast Seasonality

Use 3 years of history to time inventory builds ahead of Diwali-Christmas, the single biggest demand swing.

KEY KPIs TRACKED

Units Sold per Product · Inventory Turnover Rate · Stockout Rate · Excess Inventory Value · Seasonal Demand Patterns · Revenue Contribution per Product

3-Year Performance at a Glance



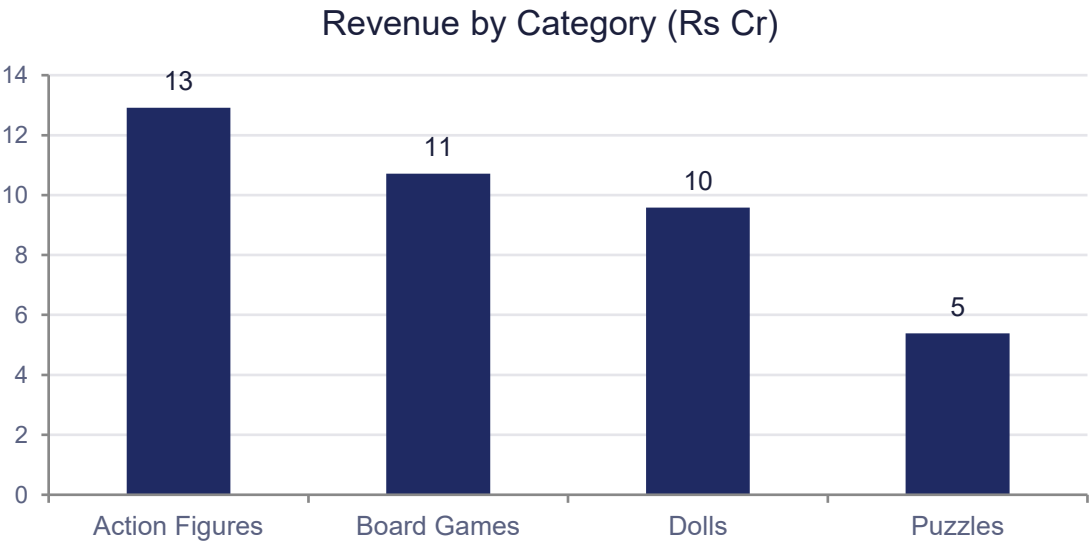
Payment / channel mix: Retail 10 stores 76% of revenue · Online 2 channels 24% of revenue | Diwali-Christmas quarter share: 39.0% of annual revenue (Oct-Dec = 25% of calendar weeks)

HEADLINE FINDING

Action Figures and Board Games together drive over 60% of revenue, but 6 high-demand products stock out roughly 17% of weeks - a direct source of lost sales. Meanwhile 6 slow-moving Dolls and Puzzles items sit at under 1x annual turnover, tying up Rs 2.26 Cr in excess inventory. The fix is smarter, product-specific reorder cadence - not blanket ordering rules.

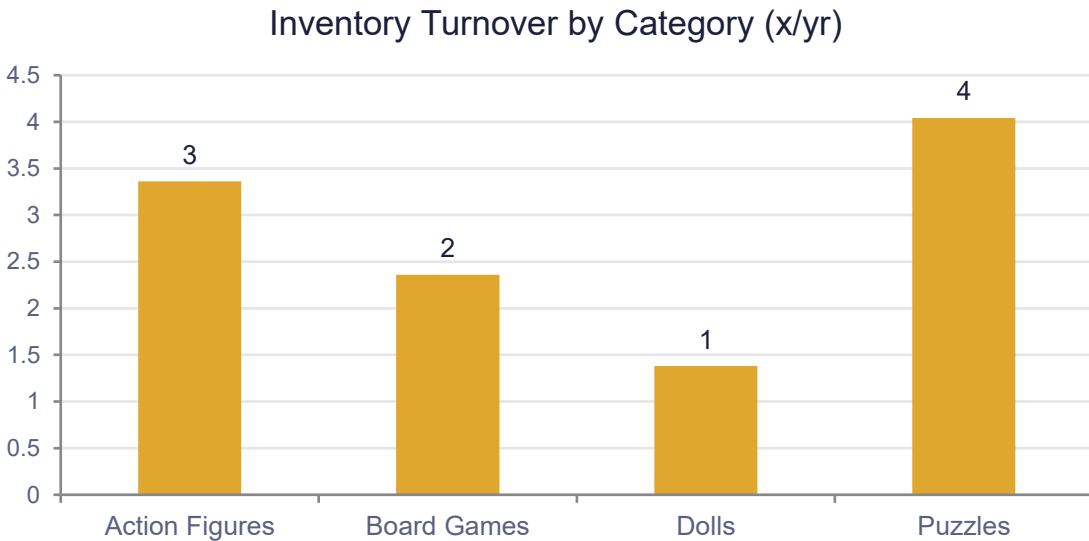
QUESTION 1

Product Category Performance



Action Figures leads revenue

Rs 12.91 Cr, 33% of total - largest category but only mid-pack 3.36x turnover.



Puzzles turn fastest

4.04x/yr turnover and 59.0% margin, the best-performing category on both counts.

Dolls carry the most risk

Only 1.38x turnover - the slowest category, and the biggest overstock contributor.

QUESTION 2 & 3

Overstock vs. Stockout: Where Capital Is at Risk

TOP 6 OVERSTOCKED PRODUCTS (LOWEST TURNOVER)

Product	Category	Turnover	Excess Value
Fairy Tale 500pc	Puzzles	0.50x	Rs 9.5 L
Baby Rosie	Dolls	0.53x	Rs 21.4 L
Fashionista Zoe	Dolls	0.54x	Rs 34.5 L
Robo Ranger	Action Fig.	0.55x	Rs 50.8 L
Trade Tycoon	Board Games	0.55x	Rs 66.3 L
Ballerina Grace	Dolls	0.56x	Rs 43.5 L

TOP 6 STOCKOUT-PRONE PRODUCTS

Product	Category	Stockout Rate	Turnover
Dino Crusher	Action Fig.	18.3%	81.3x
Super Hero Max	Action Fig.	18.3%	93.2x
Ocean Wonders 1000pc	Puzzles	18.2%	94.7x
Space Explorer 200pc	Puzzles	18.0%	92.2x
Family Fortune	Board Games	16.2%	90.1x
Doctor Dolly	Dolls	15.6%	91.1x

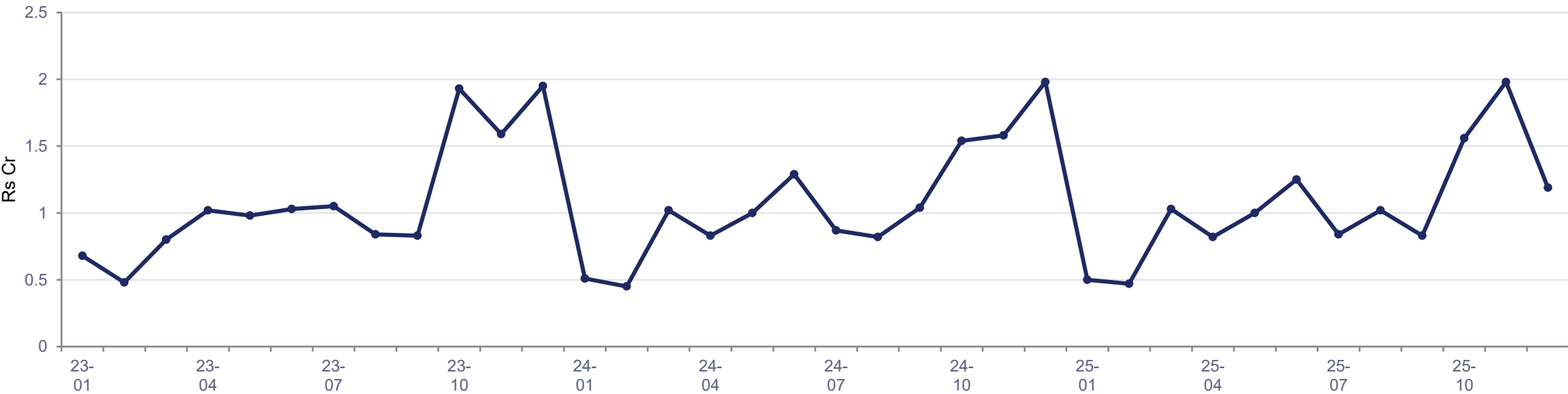
WHAT'S DRIVING THIS

Overstock (Rs 2.26 Cr tied up): 6 products - concentrated in Dolls and mid-tier Puzzles/Board Games - are reordered well above actual demand, sitting under 0.6x annual turnover.

Stockouts (~17% of weeks, 6 products): The fastest-turning items (80-95x/yr) are replenished on the same fixed cadence as everything else, so supply can't keep pace with demand spikes - especially in the festive quarter.

SEASONALITY

Monthly Revenue Trend - Jan 2023 to Dec 2025

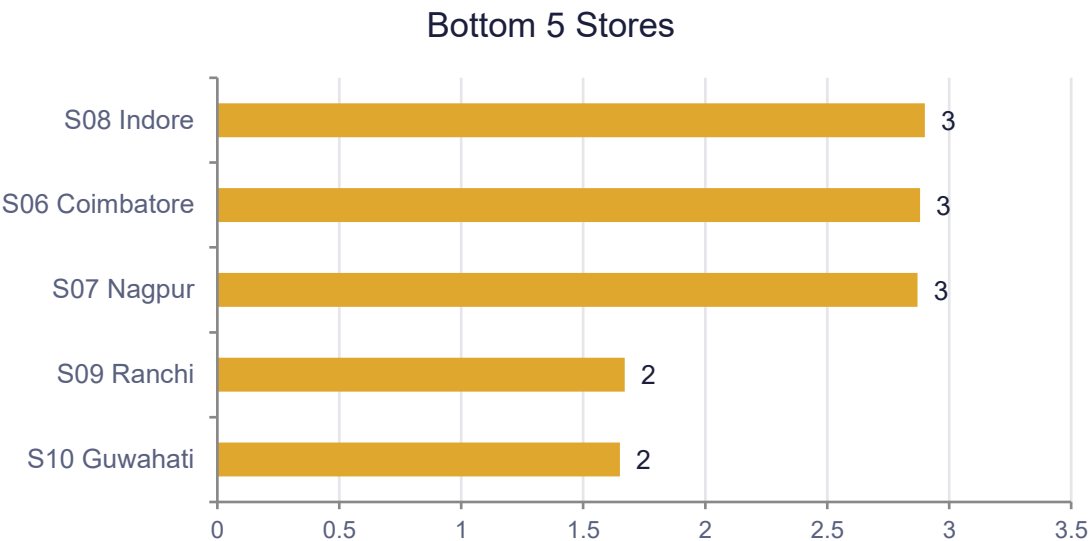
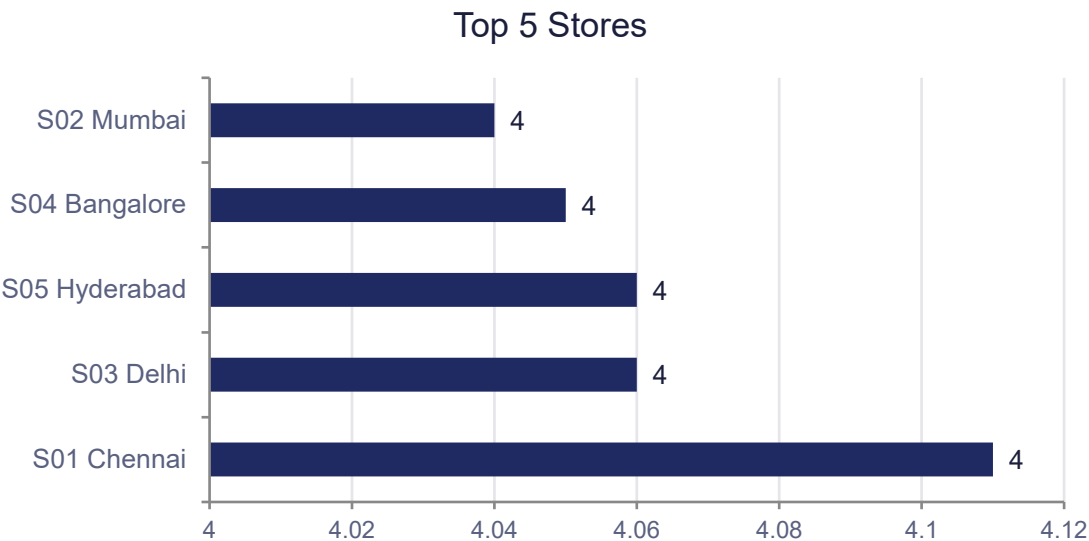


Season	Revenue	Share
Diwali-Christmas (Oct-Dec)	Rs 15.04 Cr	39.0%
Regular	Rs 14.12 Cr	36.6%
Summer / Back-to-School	Rs 6.40 Cr	16.6%
Post-Holiday Lull (Jan-Feb)	Rs 3.02 Cr	7.8%

Seasonal insight: Oct-Dec (Navratri, Diwali, Christmas) consistently peaks near Rs 1.9-2.0 Cr/month, roughly 3x the Jan-Feb trough of Rs 0.45-0.51 Cr. This is the single largest, most predictable demand swing in the business - and the primary window to time inventory builds and clearance promotions around.

QUESTION 4

Store Performance - Top 5 & Bottom 5



PERFORMANCE GAP

Top-5 Tier-1 city stores average ~Rs 4.06 Cr in revenue, nearly 2.5x the ~Rs 1.66 Cr averaged by the bottom-2 Tier-3 stores (Guwahati, Ranchi). Turnover is more even (2.2-2.5x across all stores) than revenue, meaning the gap is largely footfall and assortment depth, not inventory mismanagement at the store level - online channels (ONL1, ONL2) sit mid-pack and are still growing share.

QUESTION 5

Product Segmentation & Reorder Strategy

High-Demand Risk

6 products · 29.4% of revenue · ~90x turnover

Stock out ~17% of weeks. Strategy: raise safety stock, shorten reorder cycle to weekly, and pre-build stock 4-6 weeks ahead of Oct-Dec.

Seasonal Spike

8 products · 35.0% of revenue · ~5.1x turnover

Demand concentrated around festivals. Strategy: forecast off 3-year seasonal index, build inventory 6-8 weeks pre-Diwali, clear residual stock in Jan.

Slow Mover

7 products · 21.8% of revenue · ~1.0x turnover

Consistently overstocked, mostly Dolls & mid-tier items. Strategy: cut standing reorder quantity by 40-50%, bundle with fast movers, run clearance pricing.

Steady Seller

3 products · 13.8% of revenue · ~5.6x turnover

Predictable, healthy turnover. Strategy: keep current reorder point/quantity rules - use as the reference model for other segments.

Strategic Recommendations

QUICK WINS (0-3 MONTHS)

- Cut standing reorder quantities 40-50% on the 6 slow-moving Dolls/Puzzles items to release Rs 2.26 Cr in tied-up capital
- Move the 6 stockout-prone products to weekly (not monthly) replenishment review
- Bundle slow movers with fast movers in in-store and online promotions to clear excess stock
- Launch a clearance markdown on excess Dolls inventory ahead of the post-holiday lull

LONG-TERM (3-12 MONTHS)

- Build a seasonal demand forecast model using the 3-year Oct-Dec index to set reorder points 6-8 weeks ahead of festivals
- Set product-specific reorder points and quantities by segment (High-Demand Risk, Seasonal Spike, Slow Mover, Steady Seller) instead of one blanket policy
- Stand up a live Power BI dashboard tracking turnover, stockout rate and excess inventory value by product and store
- Invest hyperlocal marketing/assortment depth in Tier-3 stores (Guwahati, Ranchi) to close the ~2.5x revenue gap with Tier-1 stores

Assumptions & Data Notes

Synthetic dataset

45,062 weekly product-store records across 24 products, 4 categories, and 12 stores (10 retail + 2 online), Jan 2023-Dec 2025, generated to reproduce the patterns named in the case brief (overstock, stockouts, seasonality).

Missing data by design

~5% of UnitsInStock and ~1% of UnitsSold rows were blanked, and ~4% of Category labels use inconsistent spellings/casing, matching the assignment's data-cleaning requirement. Missing stock was imputed via forward/backward fill within each product-store series; missing units sold via product-store median.

Turnover & stockout definitions

Inventory Turnover = annualized units sold / average total stock on hand. Stockout weeks are those with UnitsInStock at or below 2 units at a given store.

Store tiers

5 Tier-1, 3 Tier-2, 2 Tier-3 retail stores plus 2 online channels, weighted so Tier-1 stores naturally outperform - reproducing the top/bottom-5 pattern in the brief.

Scope limits

Three years of history only; no external data sources used, per assignment constraints. Figures are illustrative for the case study, not ToyJoy's actual financials.



Thank You

ToyJoy Inventory Optimization · Data Analytics Case Study

Rs 38.58 Cr

Revenue

54.3%

Avg Margin

12

Stores

Rs 2.26 Cr

Excess Freed